

Milestone #3
Semipalmated Sandpiper (SESA) Migration Stopover Site Analysis
Carissa Rego | Bryant Summer 2026 | Capstone Project

Main Goal

Identify which habitat and land cover characteristics predict SESA stopover site usage during migration, and evaluate the long-term vulnerability of these stopover sites to sea level rise.

Hypothesis

SESA stopover site usage is predictable using land cover characteristics (particularly the presence of coastal wetland habitats). Sites that have experienced habitat degradation between the years 2001-2025 will be less likely to be used as stopover sites and therefore will have a lower chance of SESA being present.

1. Choose and Justify Model

For my base model I chose Logistic Regression because I am performing a binary classification task to determine whether or not SESA will be present a particular site. Logistic Regression is simple and interpretable with the coefficients showing both magnitude and direction of a features impact on the prediction. A logistic regression models probability of presence as a sigmoid function of a linear combination of features, probabilities are between 0-1. A threshold of 0.5 is what converts those probabilities to class labels 0 or 1. I chose to use the top 200 detections sites as stopover sites in this analysis to be sure that all locations considered to be 'stopover sites' were used heavily and consistently by SESA.

Code flow

- Data loaded from Postgres database
- Columns dropped and cleaned
- NAs filled
- Low variance features removed
- Train test split of data
- Fit Logistic regression model
- Generate probabilities
- Set threshold and produce class labels
- Evaluate using confusion matrix, AUC-ROC, precision, recall, F1

2. Data Imbalance

I had slight data imbalance in my dataset - 200 presence stopover sites and 500 pseudo-absences. I chose to use the pseudo absence approach because it is used in SDM (Species distribution modeling) and means that random background points are representing available environments that were not utilized, not that there are confirmed absences at those particular sites. I split my data into train and test (70/30 split) data prior to data manipulation so there was no risk of data leakage occurring. I avoided using accuracy alone and in addition used AUC-ROC, precision, recall, and F1.

3. Baseline Results

The baseline LR model was trained with default parameters. The accuracy of the model was 0.863, AUC-ROC was 0.848, precision was 0.818, recall was 0.632, and F1 was 0.713. After k-fold cross validation where k=5, the AUC was 0.851, SD = 0.026, sensitivity = 0.610 and specificity = 0.958. The low variation shows that the model is stable.

4. Parameter Tuning

Leaky features such as n_detections and n_years were removed from the model because they resulted in AUC = 1.0 and convergence warnings that pointed out perfect separation. After they were removed AUC decreased from 1.0 to 0.848. Low variance features were also removed for convergence issues.

5. Evaluation Metrics

The logistic regression model was compared to a naive baseline model that always picks the majority class. It improved accuracy 13.3 percentage points up from 0.730 (naive baseline). Logistic regression also improved the naive baseline AUC-ROC of 0.500 by 0.348 which proves that the model is not just randomly guessing but instead learning actual signals. Precision = 0.818 shows that when the model predicts presence of SESA it is correct in that prediction 81.8% of the time, recall = 0.632 shows that the model can catch 63.2% of true hotspots, but misses 21 out of 57 true stopover sites in the test set, and F1-score = 0.713, which balances both precision and recall. The confusion matrix shows that in total there were 146 TN, 8 FP, 21 FN, and 36 TP.

6. Critical Analysis

Logistic regression assumes the data has linear relationships, and the preference of SESA is almost definitely a non linear relationship. The logistic regression is a conservative model, because it misses 37% of actual stopover sites. 14 of the 21 FN had all zero OHE features (land cover classes) because they are the reference category that was dropped to handle multicollinearity. This is one limitation and something that I may choose to find a different strategy for as I continue my project.

7. Next Steps

As I continue I want to use probability output as a habitat suitability score, and in tandem with this I would like to test different sea level rise scenarios that are more practical in the near future (smaller scenarios) because I believe 0.5m and 1.0m were both too large of sea level rise scenarios. I want to expand my focus past only whether or not certain sites will flood, but also to how this will impact SESA migration as these current stopover sites are lost.